CHEN-YE HONG

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Research Interests

Brain–computer interfaces and human–robot interaction, with an interest in using neural, physiological, and behavioral signals to help interactive systems adapt to users' intentions and states.

Education

National Taiwan University (NTU) Sep. 2022 – Jan. 2026
B.S. in Mechanical Engineering Taipei, Taiwan
GPA: 4.16/4.30 | Graduation Rank: 3/146

National Tsing Hua University (NTHU) Sep. 2021 – Jun. 2022 (Transferred to NTU)
Undergraduate Studies in Power Mechanical Engineering Hsinchu, Taiwan
GPA: 4.16/4.30 | Department Rank: 1/116

Publications

†Equal contribution (co-first authors).

- **C.-Y. Hong**†, C.-H. Chou†, Y.-J. Wang, K.-L. Wu, S.-J. Yang, C.-H. Chang, and D.-R. Li (2026). "An experimental cadaveric limb motion platform for detecting altered mechanical responses under small-joint defect conditions." Medical Engineering & Physics, 147(9), 095006. [DOI]
- C.-Y. Hsieh†, **C.-Y. Hong**†, J.-Z. Li†, P.-C. Yang†, Y.-C. Hsu, C.-H. Tsai, C.-H. Hsu, and Y.-T. Wang (2026). "From Movement to Manipulation: Multimodal Control of a Mobile Assistive Robot." Proceedings of the 14th International Conference on Brain–Computer Interface (BCI), IEEE. (Oral presentation) [DOI]

Submitted Manuscripts

- **C.-Y. Hong**, I. C. Lee, C.-C. Deng, J. P. Hsu, and Y. T. Wang. "RehabVerse: A Virtual-Reality Multiplayer Rehabilitation Game Scored from Muscle and Brain, with Adaptive Score Manipulation." Submitted to ACM CHI 2027.

Selected Research and Projects

LLM-Driven Robot Pet with State-Dependent Interaction Aug. 2026 – Present
Independent Project | GitHub

- Developed a MuJoCo robot pet that combines VLM-derived human behavioral cues with simulated internal states to guide LLM-based behavior selection.
- Designed speed-dependent rewards for PPO locomotion to address standing-still behavior under low-speed commands, evaluating velocity tracking and stability across 3 training seeds.

RehabVerse: VR Motor Rehabilitation with Adaptive Scoring Jun. 2026 – Sep. 2026
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Investigated motor engagement during executed and imagined movements using mu-band event-related desynchronization (ERD) in 6 healthy participants.
- Developed stage-based analyses of EEG lateralization to assess responses following score boosts and led manuscript writing and figure preparation.

Multimodal Control for an Assistive Mobile Manipulator Jul. – Aug. 2025
Summer Research Intern, BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and 3D-printed the mobile base of an EEG-controlled assistive robot that completed 2 object-retrieval tasks.
- Implemented MediaPipe-based control of an open-source 6-DOF arm, using gestures to select joints and lateral hand motion to control rotation.

Additional Research Experience

VR Motor Rehabilitation with Human and AI Competition Jun. 2026 – Present
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Conducted EEG experiments using the RehabVerse platform in a two-session study involving 22 healthy participants to compare solo, human-opponent, and AI-opponent conditions during executed and imagined movements.

Eye-Tracking and EEG-Based Communication Interface for ALS Jun. 2026 – Present
Research Assistant (Full-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Designed and fabricated a six-channel EEG headset for a planned ALS communication interface combining SSVEP-based selection and eye tracking.

Human-Centric Automated Driving: VR-Based Passenger Emotion Assessment Sep. – Dec. 2025
Student Researcher (Part-time), BCI Lab, Academia Sinica | Dr. Yu-Te Wang

- Developed a Unity-based VR driving testbed to examine passenger valence and arousal across driving styles and road environments using EEG and self-reports.

Biomechanical Testing Platform for Small-Joint Evaluation Jan. 2024 – Oct. 2025
Undergraduate Researcher, SAM Lab, NTU | Prof. Dian-Ru Li

- Designed a cadaveric limb motion platform for synchronized force measurement and optical tracking to assess mechanical changes under controlled joint defects.
- Used PCA and k-means clustering to group marker trajectories and quantified regional deformation through apparent in-plane strain analysis.

Interactive VR Simulator for Cardiac Catheterization Training Jul. 2023 – Aug. 2024
Undergraduate Researcher, Advanced Medical Device Lab, NTU | Prof. Hao-Ming Hsiao

- Developed a Unity-based VR prototype for interactive guidewire positioning and balloon-driven stent expansion.

Additional Experience

Mechanical Engineering Intern, HP Inc. Jul. 2024 – Jun. 2025

- Redesigned a laptop hinge to improve thermal performance and developed a modular SSD shielding cap adopted across 3 product lines, cutting replacement time by 86% compared with similar company designs.

Depth-Camera-Based Autonomous Driving (course project) Feb. – Jun. 2025

- Trained a YOLOv8 object detector and developed a path-planning algorithm to support obstacle-aware navigation using a single depth camera.

Leadership Experience

Founder & President Sep. 2024 – Jan. 2026
NTU Mechanical Engineering International Compass

- Founded a student organisation to expand access to overseas study information and led a team of up to 22 members that conducted 30+ alumni interviews and organised forums reaching 300+ attendees.

Honors and Awards

Dean's List Award, NTU (top 5% of class) Fall 2024, Spring 2025
Academic Excellence Award, NTHU (top 5% of class) Fall 2021, Spring 2022

Technical Skills

Programming	Python, C# (Unity), MATLAB
ML & Vision	PyTorch, OpenCV, MediaPipe, YOLOv8
Robotics & AI	MuJoCo, RL (PPO), LLMs, VLMs
EEG Analysis	MNE-Python, EEGLAB
Hardware & VR	SolidWorks, Autodesk Inventor, 3D printing (FDM/SLA), Unity